You might have heard about many types of energies used in our day-today life. For all those, electron energy is the base.

3.5.3 Important Energy Bands in Solids

Though there are number of energy bands in the solids, the following are of the important ones.

1. Valence Band

The range of energies (i.e., bands) possessed by valence electron is known as valence band.

The electron in the outermost orbit of an atom is known as valence electron. In a normal atom, valence band has the electron of highest energy. This band may be completely or partially filled. For instance, in case of inert gas, the valence band is full, whereas for other material, it is only partially filled. The partially filled band can accommodate more electrons.

2. Conduction Band

In certain material (e.g. metals) the electrons are loosely attached to the nucleus. Even at ordinary temperature, some of the valence electron may get detached to become free electrons. In fact, these free electrons are responsible for conduction of current in a conductor. For this reason, they are called conduction electrons.

The range of energy (i.e., band) possessed by the conduction electron is known as conduction band.

All electrons in the conduction band are free electrons. If a substance has empty conduction band, it means current conduction is not possible. Generally, insulators have empty conduction band. On the other hand, it is partially filled for conductors.

3. Forbidden Energy Gap

The energy gap between conduction band and the valence band on the energy level diagram is known as forbidden energy gap.

No electron of a solid can stay in a forbidden energy gap as there is no allowed energy state in this region. The width of the forbidden energy gap is a measure of bondage of valence electrons to the atom. The greater the energy gap, more tightly the valence electron are bound to the nucleus. In order to push an electron from the valence band to conduction band (i.e., to make the valence electron free), external energy must be supplied equal to the forbidden energy gap.

a) **Conductors:** Metals (e.g. Copper, Aluminium) or conductors allow the passage of electric current through them, because of large number of free electrons available in a conductor. In terms of energy band, the valence and conduction bands overlap each other as shown in Figure3.14.

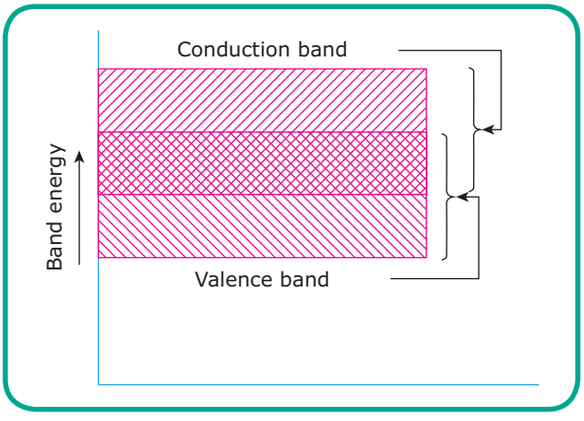


Figure3.14 Energy band of conductors

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